```
PASS 1 [2.1] 157/50 < pi
PASS 2 [2.1] pi < 22/7
PASS 3 [2.1] 41/25 < (157/50)^2/6
PASS 4 [2.1] (22/7)^2/6 < 33/20
PASS 5 [2.1] (64/25)^2 < 4*(41/25)
PASS 6 [2.1] 4*(33/20) < (18/7)^2
PASS 7 [2.84] 330000 < 10^6/3 - 5
PASS 8 [2.84] 3*s + 15 <= 4*s for s >= 330000
PASS 9 [2.84] 570^2 < 330000
PASS 10 [2.84] 330000 < 575^2
PASS 11 [2.84] sqrt(10^6) = 1000
PASS 12 [2.86] s - 4 - 1/24 > 9*s/10
PASS 13 [2.86] 9*s/10 > 100
PASS 14 [2.86] t - 2 - 1/24 > s at the smallest m
PASS 15 [2.87] smallest denominator argument exceeds numerator arguments
PASS 16 [2.87] 3/(s-5) <= 4/s
PASS 17 [2.87] 1/sqrt(1+2/s) >= 1-1/s
PASS 18 [2.87] 32/25 - 32/(25*330000) - 4/570 > 127/100
PASS 19 [2.88] 4*(s-5)^(-3/2) < 5*s^(-3/2)
PASS 20 [2.88] (127/100)^2 - 5/570 > 8/5
PASS 21 [2.90] (9/7)*sqrt(10/9) < 34/25
PASS 22 [2.90] numerator lower endpoint exceeds 127/100
PASS 23 [2.91] sqrt(2*s+10) < (283/200)*sqrt(s)
PASS 24 [2.91] 256/283 - 3/(2*570) > 9/10
PASS 25 [2.92] -127*r/100 + 9/7 < 0 for r >= 2
PASS 26 [2.92] 34*r/25 - 9/10 > 0 for r >= 2
PASS 27 [2.93] 2 + 16/570 < 3
PASS 28 [2.93] 1 + 1/r^2 <= 5/4 for r >= 2
PASS 29 [2.93] 15/(4*570) < 1/150
PASS 30 [2.95] exponential lower argument is positive
PASS 31 [2.95] lower argument = 50673/20000
PASS 32 [2.95] S_10(50673/20000) > 25/2
PASS 33 [2.95] 0 < theta=2/25 < 1
PASS 34 [2.97] first geometric moment formula
PASS 35 [2.97] second geometric moment formula
PASS 36 [2.97] third geometric moment formula
VALUE [2.97] moments=142/529,10182/12167,739822/279841
PASS 37 [2.96] curvature budget equals 3228244387/2098807500
PASS 38 [2.96] curvature budget < 31/20
VALUE [2.96] curvature_budget=3228244387/2098807500
PASS 39 [2.95] later-term mass = 71/529
PASS 40 [2.99] 0 < 91/(25*570) < 1/2
PASS 41 [2.99] exp(z) <= 1/(1-z) < 2
PASS 42 [2.99] smooth bulk mass < 3
PASS 43 [2.99] 1600*330000*575 < 10^24; used with S_50(57)
PASS 44 [2.99] s^(3/2)*exp(-sqrt(s)) is decreasing
PASS 45 [2.100] numerator replacement coefficient = 192
PASS 46 [2.100] constant coefficient in bulk exponent is below -5/4
PASS 47 [2.100] 1/s coefficient in bulk exponent is below 27/2
PASS 48 [2.100] -5/4 + 27/(2*s) < -1
PASS 49 [2.100] bulk numerator coefficient = 768
PASS 50 [2.100] bulk denominator coefficient = 4800
PASS 51 [2.100] bulk value error coefficient < 6000
PASS 52 [2.100] second-difference error coefficient = 24000
PASS 53 [2.101] 6001 < 10^6/150 - 1/10
PASS 54 [2.101] derivative of n/150 - 1/10 - 6*sqrt(n) - 1 is positive
PASS 55 [2.101] 9/10 + 1/50 = 23/25
PASS 56 [2.102] sqrt(s-4) > (9999/10000)*sqrt(s)
PASS 57 [2.102] sqrt(23*s/25) < (9592/10000)*sqrt(s)
PASS 58 [2.102] (64/25)*(9999-9592)/10000 > 1/10
PASS 59 [2.102] nonbulk value coefficient = 96
PASS 60 [2.102] nonbulk second-difference coefficient = 384
PASS 61 [2.102] 24000 + 384 < 25000
PASS 62 [2.102] s^3*exp(-sqrt(s)/10) is decreasing
PASS 63 [2.102] 570/10 = 57
PASS 64 [2.102] S_50(57) > 10^24
PASS 65 [2.102] 25000*330000^3/10^24 < 1/1000
VALUE [2.102] endpoint_error_upper=35937/40000000
PASS 66 [2.102] 8/5 - 31/20 - 1/1000 = 49/1000
PASS 67 [2.102] final curvature margin > 0
VALUE [2.102] final_margin=49/1000
COMPLETE_PASS checks=67 failed=0 arithmetic=GMP_exact_rational elapsed_us=206
```